Public Health & Epidemiology

A comprehensive overview of population health, disease prevention, and the social factors that shape our well-being.

Course: NURS4324

Level: 4

Credit Hours: 3

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Global Health Challenges: Dual Threats

In today’s global context, human health faces multiple complex threats.

Communicable Diseases (CDs)

Diseases like **AIDS, tuberculosis, and malaria** continue to cause high morbidity.

Particularly devastating in low- and middle-income countries.

Emerging pathogens highlight the urgent need for surveillance.

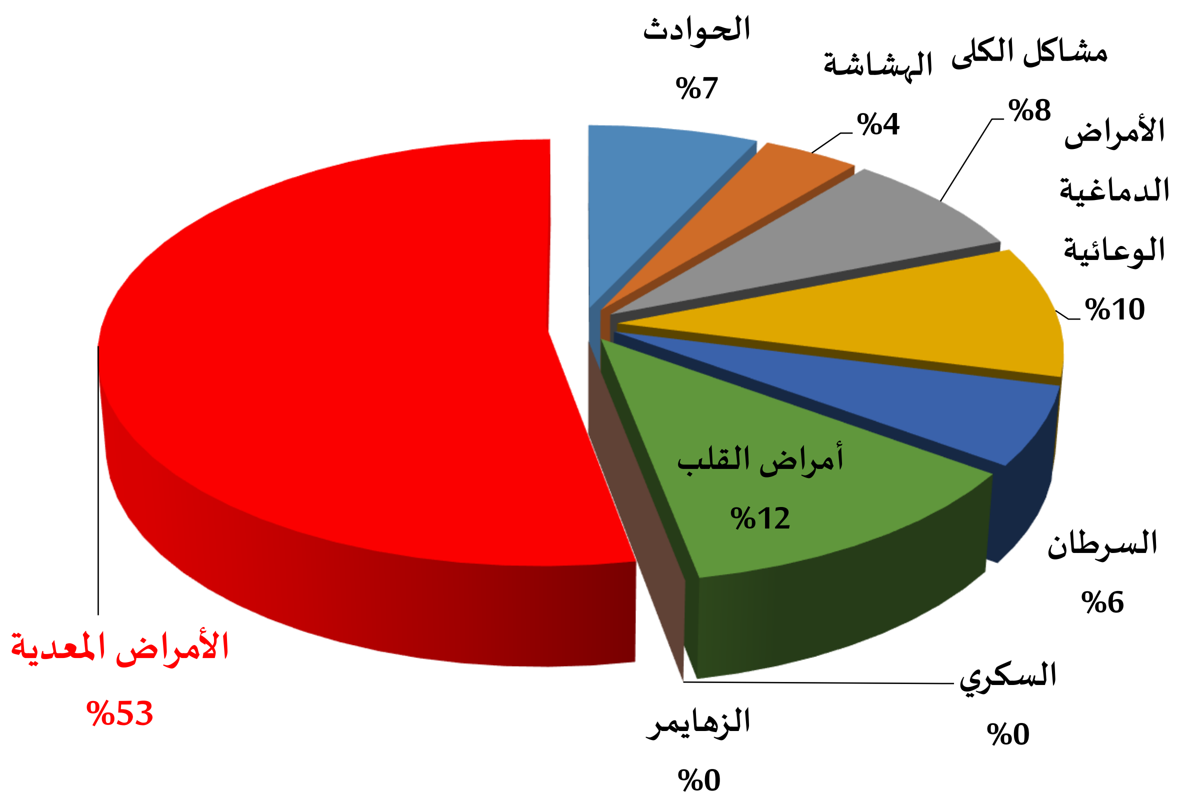
Non-Communicable Diseases (NCDs)

36 million deaths annually are attributed to NCDs worldwide.

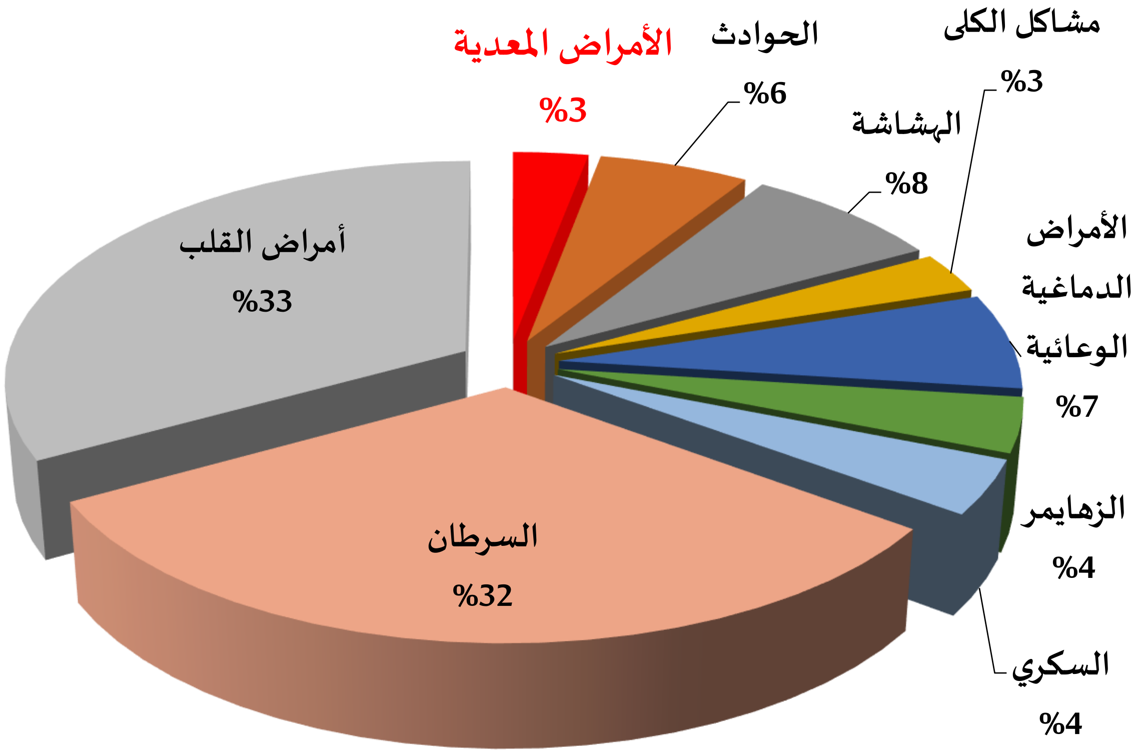
Cardiovascular disease, cancer, and diabetes are leading leading causes of death.

NCDs account for nearly **80% of all deaths** in major groups.

مسببات الوفاة في الولايات المتحدة الأمريكية - 1990



مسببات الوفاة في الولايات المتحدة الأمريكية - 2010



Limitations of Medical Treatment

Medical treatment alone is not always effective for controlling these diseases

High cost of medical care and shortage of facilities in low- and middle-income countries

Financial barrier: average NCD treatment cost exceeds 40% of annual household income in LMICs

Systemic gap: <30% of rural populations in LMICs have access to essential NCD services

Defining Public Health: A Core Mission

Understanding the broad mission and diverse definitions of public health.

Core Mission and WHO Definition

Core Mission: Fulfilling society's interest in assuring conditions in which people can be healthy.

WHO Definition: All organized measures (public or private) private) to prevent disease, promote health, and prolong prolong life among the population as a whole.

UK Perspective and IPC Integration

UK Perspective: The art and science of preventing disease, disease, promoting health, and prolonging life through through the organized efforts of society.

IPC Integration: Public health aims to prevent transmission, protect healthcare workers, and respond respond rapidly to outbreaks.

Basic Functions of Public Health

Health promotion and protection

Empowering individuals and communities to adopt healthy behaviors and safeguard well-being

Health assessment

Systematic collection, analysis, and interpretation of health data to inform policy and practice

Disease prevention

Implementing evidence-based interventions to reduce incidence and severity of disease

Disease surveillance

Ongoing, systematic collection and analysis of health-related data for early detection and response

These four core functions form the foundation of modern public health practice

Public Health

Collective actions to improve population health

Elements of Modern Public Health

Collective responsibility

Shared accountability across sectors, institutions, and communities for health outcomes

Prime role of the state in protecting and promoting public health

Policy leadership, regulation, financing, and stewardship of health systems

Partnership with the population served

Co-creation of programs, participatory decision-making, and community ownership

Emphasis on prevention

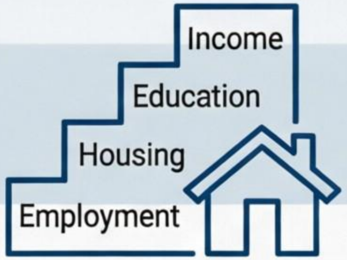
Primary, secondary, and tertiary prevention as core operational principle

These four elements constitute the Beaglehole and Bonita (1997) framework for modern public health practice

Additional Modern Public Health Elements

Recognizing underlying socio-economic determinants of health and disease

Integration of evidence-based practice and community engagement



“Underlying socio-economic determinants of health and disease”

“Evidence-based practice and community engagement”

Public Health Disciplines Overview

Family and community medicine specialists

Deliver primary care, coordinate health services, and bridge clinical and population-level practice.

Health promoters and educators

Design and implement health communication campaigns and behavior-change interventions.

Public health nurses

Provide community-based care, conduct home visits, and lead maternal-child health programs.

Public health informatics

Develop and manage health information systems, surveillance platforms, and data analytics infrastructure.

Infection control officers

Prevent healthcare-associated infections and support outbreak response in institutional settings.

Public health nutritionists

Assess dietary patterns, design food policy interventions, and address malnutrition across life course.

Policy analysts

Evaluate health legislation, model policy impacts, and advise on regulatory frameworks and resource allocation.

Epidemiologists

Investigate disease distribution, identify risk factors, and quantify burden through rigorous study designs.

Environmental health officers

Monitor air/water/soil quality, enforce health regulations, and mitigate environmental hazards to human health.

These disciplines collectively form the interdisciplinary workforce essential for effective public health action

who is responsible for public health?

all sectors of society

Health departments

coordinated efforts - other stakeholders

Other areas of government

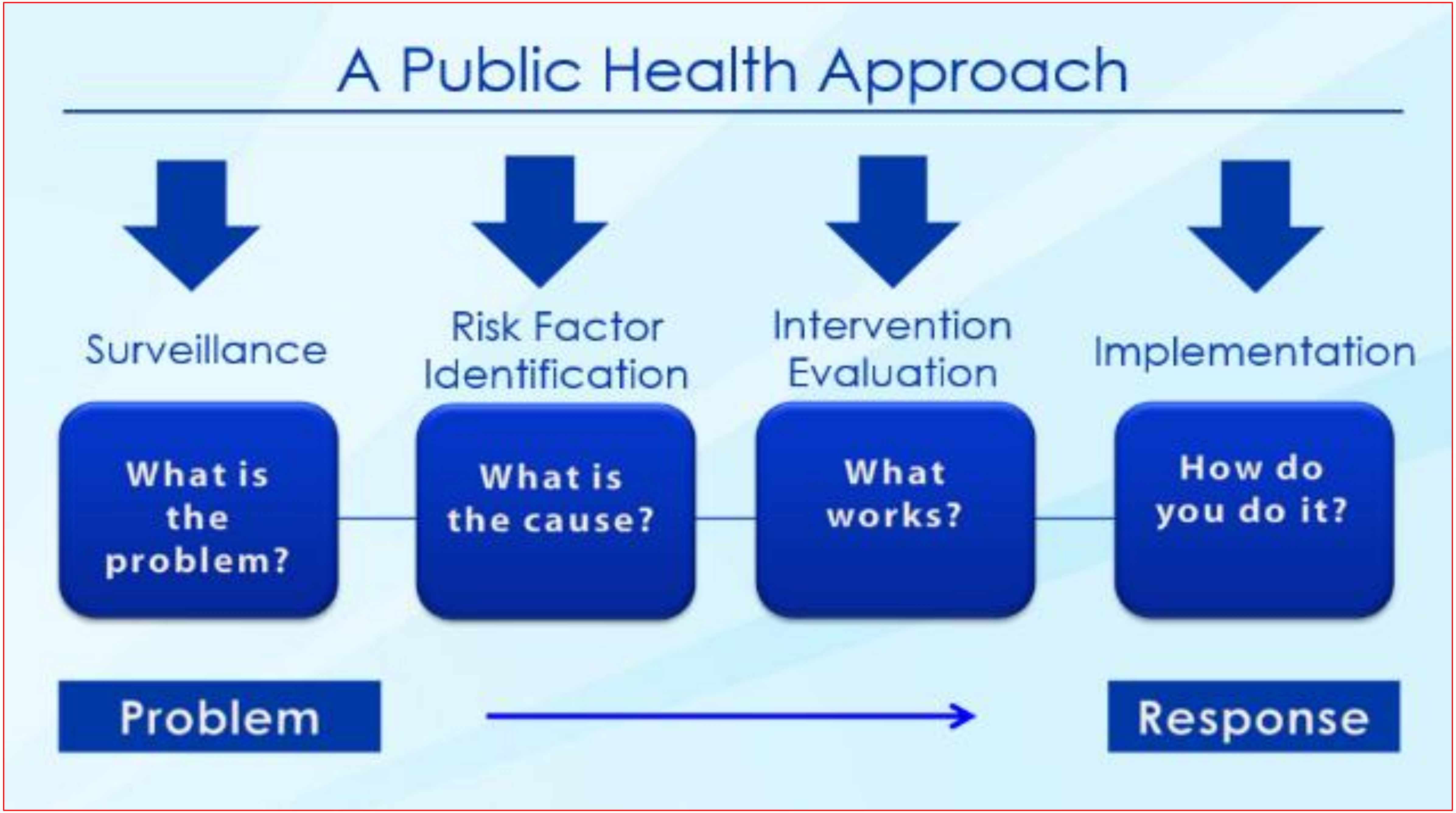
Community

Private sector

NGOs

International Organizations

Improving
Health



A Public Health Approach

Surveillance

What is the problem?

Risk Factor Identification

What is the cause?

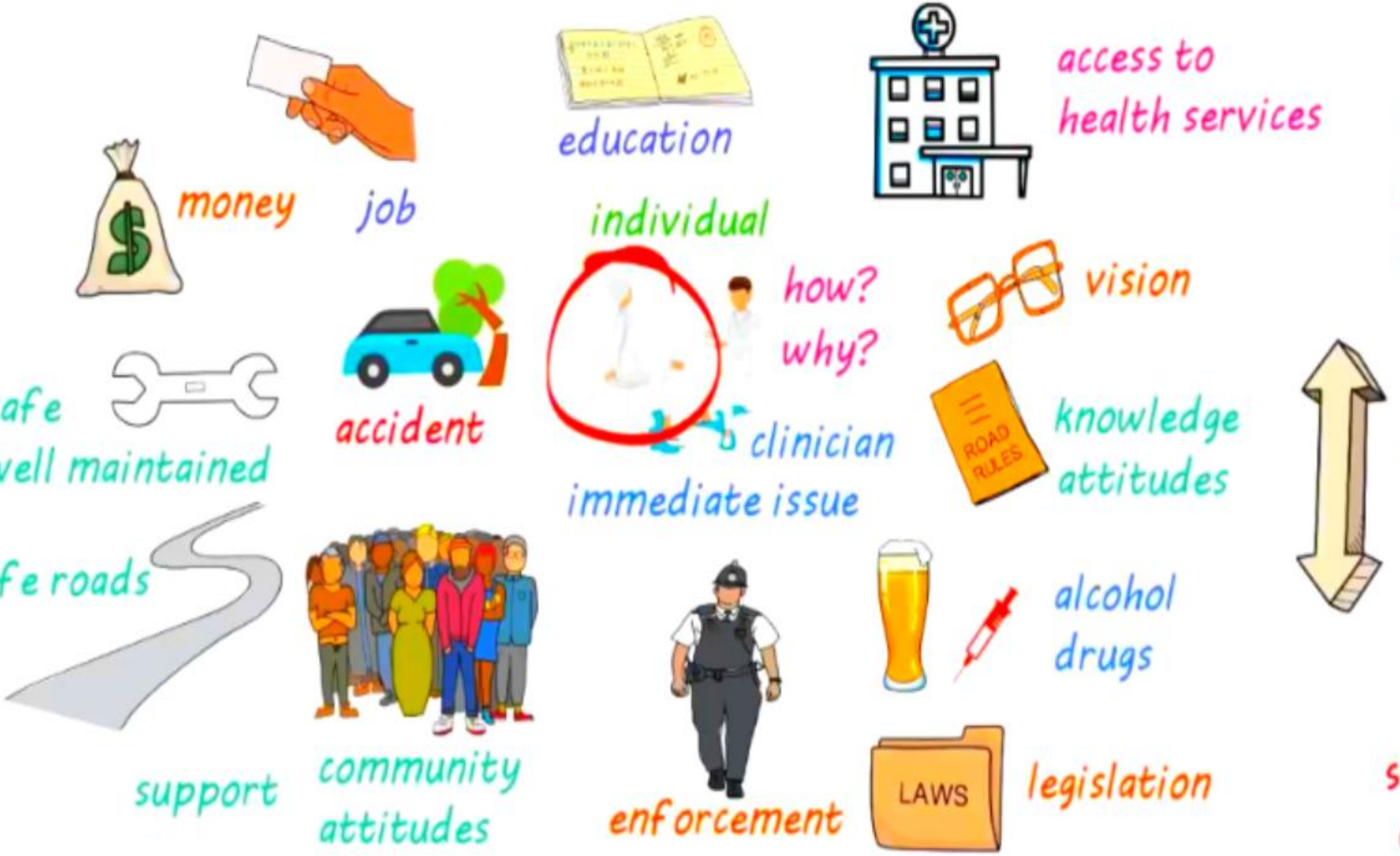
Intervention Evaluation

What works?

Implementation

How do you do it?

Problem → Response



money

job

education

individual

access to health services

safe

well maintained

accident

how?

why?

clinician

immediate issue

vision

knowledge

attitudes

safe roads

support

community attitudes

enforcement

alcohol

drugs

legislation

complex interaction

determinants of health

individual

lifestyles and behaviour

physical, social and economic environment

social determinants of health

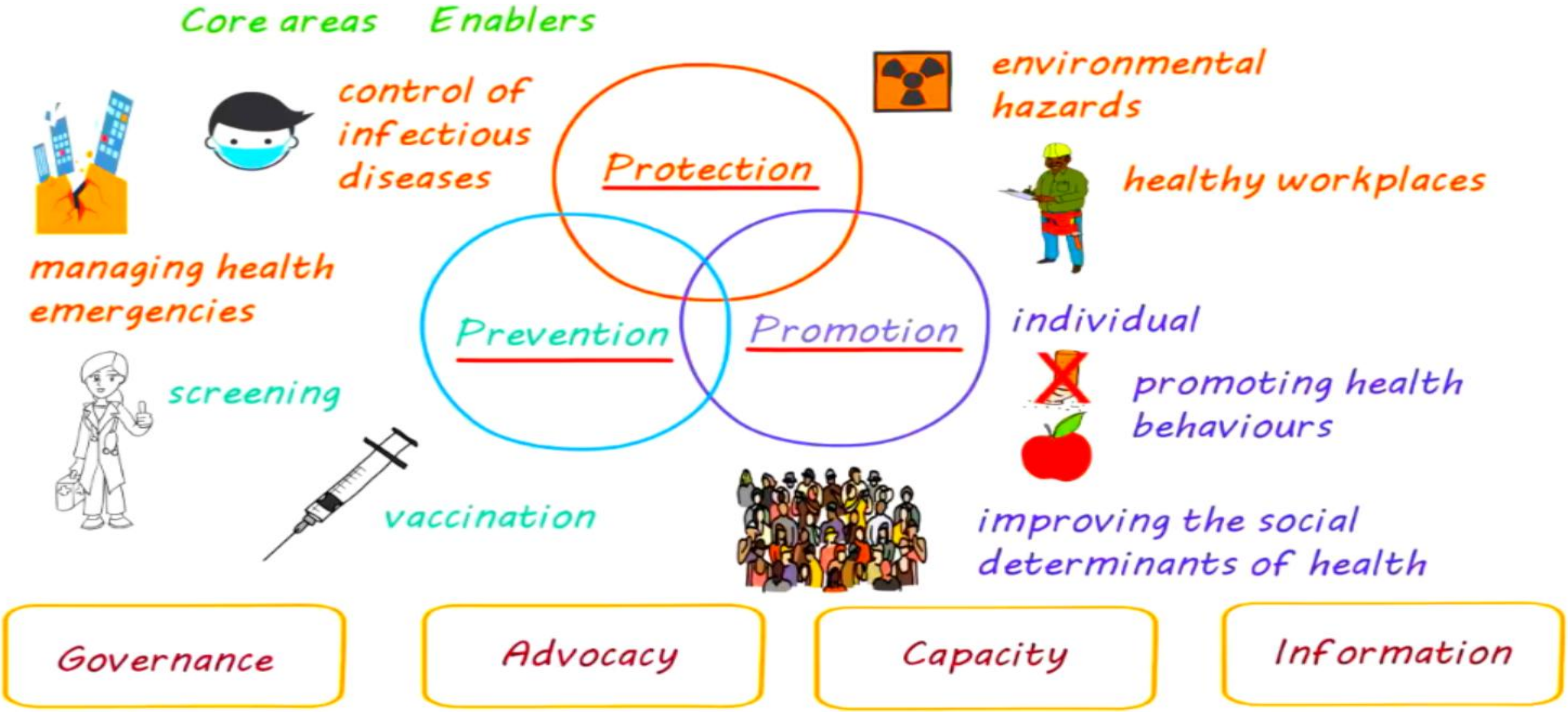
public health → benefits entire population

upstream factors

Medicine vs Public Health Comparison

Medicine	Public Health
Concern: Individual	**Concern:** Community
Focus: Treating diseases	**Focus:** Preventing disease and promoting good health
Responsibility: Patient freed of responsibility	**Responsibility:** Community members participate in solutions
Basic Sciences: Anatomy, physiology, pathology	**Basic Sciences:** Biostatistics, epidemiology, social sciences
Priority: Helping one patient regardless of cost	**Priority:** Benefiting large numbers in community
Cost: High	**Cost:** Low
Technology: Usually sophisticated	**Technology:** Simple

How does public health work?


Core areas
Enablers
control of infectious diseases
managing health emergencies
Protection
environmental hazards
healthy workplaces
Prevention
screening
vaccination
Promotion
individual
promoting health behaviours
improving the social determinants of health
Governance
Advocacy
Capacity
Information

Achievements of Public Health

- Vaccination
- Safer Workplaces
- Safer & Healthier Food
- Motor Vehicle Safety
- Recognition of Tobacco Use as a Health Hazard
- Healthier Mothers and Babies
- Fluoridation of Drinking Water

Health Definition: WHO 1948

A state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity

"Physical"

"Mental"

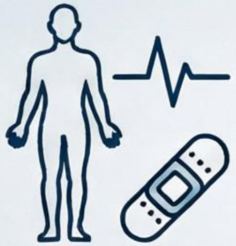
"Social"

"Spiritual"

"Emotional"

Health Dimensions and Key Determinants

Physical Health: Absence of disease and injury, affects other health dimensions
Mental Health: Ability to think and learn, good self control, self esteem, problem solving
Social Health: Ability to make and maintain meaningful relationships and interactions
Emotional Health: Expressing emotions appropriately, maintaining integrity under stress
Spiritual Health: Sense of purpose, integrity, principles, ethics, commitment to higher values


"Physical Health"

"Mental Health"

"Social Health"

"Emotional Health"

Purpose
Purpose
Ethics
Values
"Spiritual Health"

"Income"

"Employment"

"Education"

"Social and physical environments"

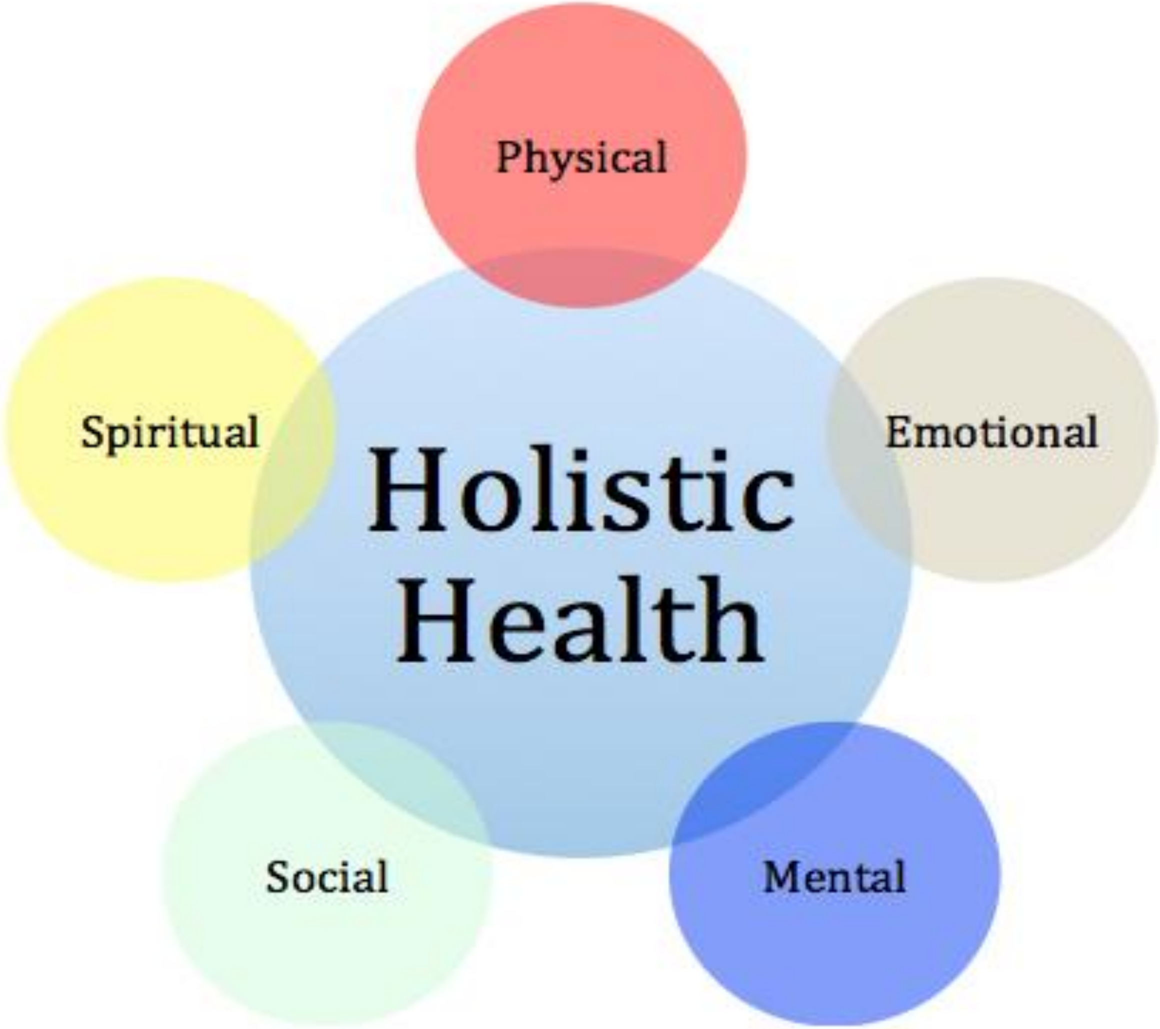
"Child development

"Personal practices"

"Health services"

"Social support"

"Biology, gender, culture"



Physical

Spiritual

Emotional

Holistic Health

Social

Mental

The 12 Core Determinants of Health

A comprehensive framework for understanding population health outcomes.

1 Income & Status

Prosperity and equitable wealth distribution are primary health drivers.

2 Employment

Control over work conditions and lower stress lead to longer longer lives.

3 Education

Higher education correlates with better health and lower stress stress levels.

4 Social Environments

Values, safety, and cohesive communities reduce health risks. risks.

5 Physical Environments

Quality of air, water, housing, and workplace safety standards. standards.

6 Child Development

Early childhood experiences shape lifelong health and coping coping skills.

7 Personal Practices

Lifestyle choices like nutrition, activity, and avoiding tobacco. tobacco.

8 Social Support

Networks of family and friends act as buffers against life's stress. stress.

9 Biology & Genetics

Inherited traits and predisposition to specific health conditions. conditions.

10 Health Services

Access to effective preventive and curative medical care.

11 Gender

Different health risks and needs across the gender spectrum. spectrum.

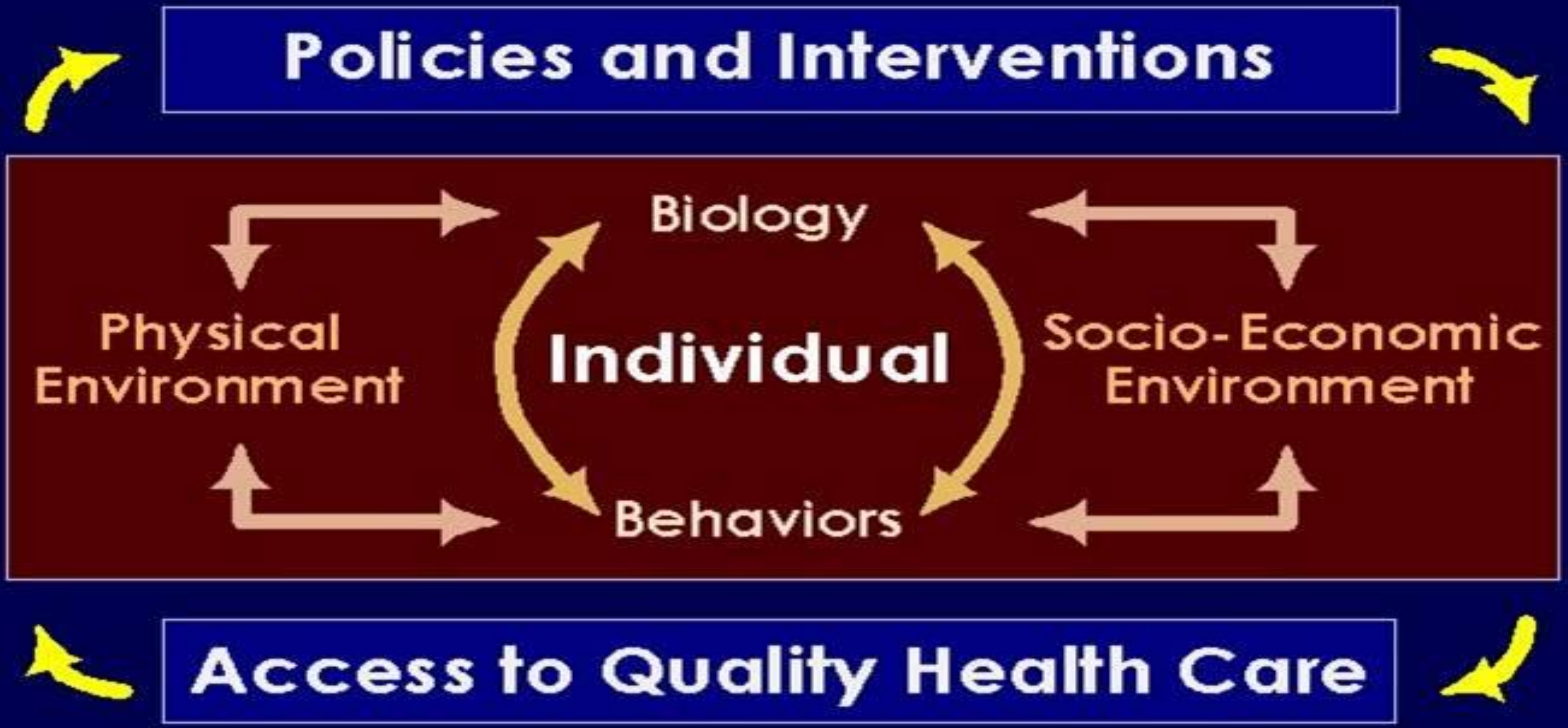
12 Culture

Beliefs, traditions, and norms that influence health behaviors. behaviors.

Determinants of Health

- Income and Social Status
- Social Support Networks
- Education
- Employment and Working Conditions
- Physical Environments
- Biology and Genetics
- Personal Health Practices and Coping Skills
- Healthy Child Development
- Health Services and Social Services
- Social Environments
- Gender
- Culture

Determinants of Health



Policies and Interventions

Biology

Physical Environment

Individual

Socio-Economic Environment

Behaviors

Access to Quality Health Care

Healthy People 2010

Thank you for your attention!